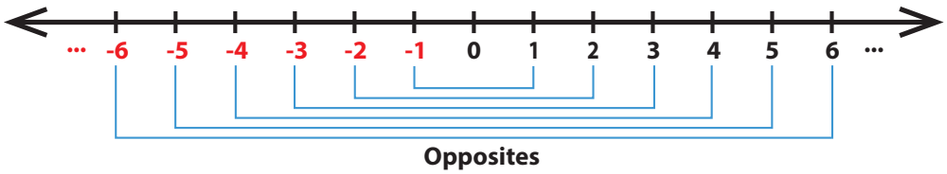
the number line. That is, the rabbits R and S stand at the same distance from 0, but in opposite directions.

Here, 6 and −6 are ***opposite*** to each other. That is, two numbers that are at the same distance from 0 on the number line, but are on the opposite sides of it, are ***opposite*** to each other. For every positive integer, there is a corresponding negative integer and vice versa. The opposite of each integer is shown in the figure.



Note

The opposite of the opposite of a number is the number itself. For example −(−5) read as negative of negative 5 or minus of minus 5 is equal to 5 itself.

Now, it is easy to write the opposite of the numbers −7, 12, −225 and 6000. Note that, the opposite of a positive integer is negative, and the opposite of a negative integer is positive, whereas the opposite of zero is zero.

Number	Its opposite
12 or +12	−12
−7	+7 or − (−7) = 7
−225	+225
6000 or +6000	−6000

The "opposites" are naturally more convenient to relate and understand with many of our daily-life situations like saving-spending, credit-debit, height above-below, where

i) the saving is treated as positive and the spending is treated as negative.

ii) a credit is considered positive whereas a debit is considered negative.

iii) the height above the sea level is regarded as positive and the height below the sea level is regarded as negative.

Example 2 Represent the following situations as integers:

i) A gain of ₹1000 ii) 20°C below 0°C iii) 1990 BC (BCE)

iv) A deposit of ₹15847 v) 10 kg below normal weight

Solution

i) As gain is positive, ₹1000 is denoted as + ₹1000.

ii) 20°C below 0°C is denoted as −20°C.

iii) A year in BC (BCE) can be considered as a negative number and a year in AD (CE) can be considered as a positive number. Hence, 1990 BC (BCE) can be represented as −1990.

iv) A deposit of ₹15847 is denoted as + ₹15847.

v) 10 kg below normal weight is denoted as −10 kg.

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